

Institute of Actuaries of India

Subject ST1 – Health and Care Insurance Specialist Technical

March 2017 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

- Protecting the nation's health
- Safeguard healthy workforce
- Improve productivity and GDP
- Redistribution of wealth
 - Ensure that the poorest members of society have access to medical care.
 - Since funded by increase in income tax, benefit will be funded by rich for poor
- Political considerations / social culture
Health provision is an emotive issue and likely to be politicized
- Government may have made electoral promises
- Ensure adequate provision of services to the elderly and for children

Reasons for earmarked tax levy

- Equity: higher earners pay more
- Ensure that high costs of healthcare provision are borne by whole economy rather than those in need.

(4)

(ii)

- Introduce mandatory private provision
 - For certain types of procedure/health provision, it should be ensured that mandatory requirement to provide for them is there
 - Especially for those above a certain income threshold
- Incentivise private provision
- Restrict public provision and encourage a private healthcare market to act as a top-up
- Offer tax incentives for effecting private healthcare e.g. to individuals, employers
- Provide a subsidy to private healthcare providers
- Require co-payment

(3)**[7 Marks]****Solution 2:**

Premium reduction mechanism

General point: Reducing premiums will reduce profit per policy unless accompanied by actual change in practice of company

1. Target business which poses lower risk in terms of morbidity to allow for optimistic assumption in this respect
Age / gender/ geography / lifestyle / class of product (standalone / rider etc.)
/distribution source / occupation class

Risk:

- Complicated approach which might increase sales process leading to loss of new business
2. Implement stricter underwriting procedures to ensure better classification of risks and parity with pricing assumptions

Risk:

- May increase number of policies rejected on underwriting grounds defeating the purpose of reducing premiums
 - Lengthen sales processes.
 - Increased cost of underwriting may outweigh benefits of better risk targeting
3. Reduce anti selection effect for all types of products / over-insurance cap replacement ratio in case of IP
 Reduce anti selection effect especially by catching multiple policies
 Reduce over insurance effect by ensuring regular policy reviews
 Better training of sales teams to reduce anti selection effects / over insurance

Risk:

- Introduces additional costs into process costs may outweigh benefits
 - May not meet customers' needs
4. Reduce claims outgo
 Reduce claims incidence for a given risk depending the type of product considered
 Increase exclusions applied
 Implement more stringent claims control processes
 Change basis of claims payment for each product type. E.g. from own occupation to own/suited or to any occupation for IP, specifying maximum amount of procedures for CI, specifying hospitals in case of PMI

Risk:

- This would lead to decline of more claims leading to Reputational risk
 - PRE considerations if claims philosophy not in line with policyholders expectations
5. Reduce claims severity
 Increase deferred period
 Introduce linked claims clause to encourage return to work
 Offer lower expiry age / term
 Reduce escalation of claims paid
 Limit duration of claims payment
 Implement rehabilitation services to help claimants back to work
 6. Reduce expense loadings
 Streamline processes to enable economies of scale
 Redistribute expenses between policies (e.g. apply using percentage of premium basis so that higher value policies carry higher expenses).

Allocation between policy sizes. (e.g. risk that larger number of smaller premium policies sold and expenses not covered).

7. Reduce profitability loading

Profit can be fixed in money terms so that increased volumes leads to same amount of total profit in spite of per policy lower profits

Risks:

- Elasticity of demand not accurately understood. Increases in volume not as expected.
- Competitor respond by lowering premiums further leading to price war which does not result in volumes increasing as expected.

8. Enhance retention and reduce lapse assumption

Implement a no claims discount

Risk:

- This might lead to High lapse rates following claims

9. Pay less commission

Risk:

- Insufficient commission to incentivise sales

10. Investment income

Increase investment income from held funds

Risk:

- Investment return mismatch risk if deviate from fixed interest securities
- Need to consider whether any regulatory risk associated with mismatch

11. Reinsurance

Make effective use of reinsurance arrangements

Make use of tax / solvency arbitrage

Use data available with reinsurer

Risk:

- Default risk
- Reinsurer would provide rates based on truncated data

12. Guarantees

Remove/reduce the level of guarantees in the policy

Introduce or extend use of reviewable/renewable premiums

Move to unit-linked format

Risk:

- Might make product unattractive specially if similar products offers guarantees

13. Move to ULIP format

Admin costs likely to increase significantly

[15+5=20 Marks]

Solution 3:

i)

Claims are normally assessed in yearly cohorts. It may also be done quarterly for up to date position on below:

- Check costs by provider (for heavy chargers)
- Check preferred providers, service level agreements
- Changes in practices at u/w, claims stages
- Changes in distribution methods
- Changes in terms and conditions

Key indices are:

1. Rate of claims incidence
 - No of claims / exposed to risk
 - Split by procedure/benefit types
 - Compare actual v expected
 - Split by occupation or other rating factors
 - Check cost of 'family' claims against assumptions
2. Average cost per claim
 - Total incurred claims / no of separate claims
3. Claims expenses
4. Loss ratio
 - Total incurred cost of claims / relevant earned premium

Calculated for cohort of business under review

Consider cohort by duration in light of selection effect

But depends on type/extent of medical underwriting at outset

Incurred claims include:

- Claims paid to date
- Reported but not paid
- Incurred but not reported

(3)

ii)

Additional claims factors which might be considered if claims analysis were to be used as part of a pricing calculation might include:

Trends in costs of procedures, drugs and dressings

Overall private hospital capacity, current and projected

Competitors' rates movements where these are caused by their own claims experience

(1)

iii)

Own claims data is likely to be more relevant but may not be available as this is a new product.

Since own data is not available, data from other sources can be used:

- Professional/governmental bodies
- Other countries
- Reinsurers

Other measures are as below:

1. Implement benefit caps to limit claims per case and hence maximum liability
2. Benchmark premiums against any similar plans already in the market
3. Build margins into pricing and reserving to reflect the uncertainty
4. Avoid guarantees, both for premium and benefit
5. Write product on a short-term basis without guaranteed renewability
6. Reinsure appropriately to limit retained risk
7. Ensure adequate capital is in place to meet uncertain claims outgo
8. Collect relevant data from policies written to accumulate own experience as quickly as possible
9. Delay product launch until better data is available
10. Tight benefit conditions
11. Claims preauthorisation
12. Monitor claims experience
13. Excesses
14. Preferred provider acupuncturists
15. Introduce NCDs
16. Offer cash benefit, not indemnity

(8)

iv)

Increased Anti selection

- Individuals may have information not available to insurers and hence such individuals are more likely to insure which may Change claims frequency of portfolio due to higher number of claims
- Or such individuals may insure for more amount on average

Overall, increases average cost of claim

Lack of affordability

- On account of anti-selection, Premiums might be loaded for extra risk
- This might make premiums unaffordable
- Individuals may become uninsurable
- More common to exclude certain conditions within PMI context

Availability and cheapness of tests

- However, If tests become sufficiently cheap may enable much better targeting of premium to
- Risk by using in the underwriting process.
- Depends on the costs saved in terms of claims cost against the cost of the tests

(4)

[16 Marks]

Solution 4:

Direct sales force — Face to face

Usually employed staff of the company

Cost:

- Commission payments
- Salary
- Combination
- Direct sales force – expenses (e.g. office space, cars)
- Training costs

Direct sales — Telephone selling, Internet

- For telephone selling, remuneration as above
- Phone or internet bill
- For internet selling, cost of developing and maintaining software
- Cost of web aggregators

Direct Marketing

Mail shot

Press advertising / return coupon

Cost:

- Costs of campaigns / expense of contact
- Advertising costs

Common costs:

- Head office / overheads
- Compliance costs
- Training Costs

[7 Marks]

Solution 5:

i)

Some of the possible data sources include:

Own Data - Most relevant and accurate

Issues

- Small sized company so volume of own data available limited, less credible
- No experience in CI plans. PMI data may not be relevant for pricing CI as illness covered under PMI and CI are usually different.

Population data - Data published usually by government agencies, important source with large volume

Issues

- Accuracy and reliability is questionable
- Data may not always be available in electronic format
- Experience may be out-of-date
- Morbidity experience of population data may be significantly different from insured lives data

Reinsurer data – Large volume, Accurate and based on insured lives.

Issues

- Usually charged by Reinsurer for providing data assistance
- May not be completely relevant as underwriting and claims management of the company may differ

Market data – Insured lives data, combined data of all insurers published for benefit of all.

Issues

- May take time to publish updated data
- May not be completely relevant as claims and underwriting of each company may differ

Other data – Overseas data, consultant's data, trade magazines etc.

Issues

- May not be relevant to local country
 - Consultants may charge a fee for sharing data
 - Magazine data may not have all the necessary details required for the purpose of pricing
- (5)**

- ii) Per policy reserve – typically calculated as discounted value of expected income and outgoes
- Unearned premium reserve – premium received in respect of period for which risk has not expired
 - Unexpired risk reserve – reserve in respect of unexpired risk where premium basis is inadequate
 - Incurred but not yet reported – reserve for claims which have incurred but not yet reported to the insurer
 - Claims in transit – reserve in respect of claims reported but not assessed
 - Outstanding claim reserve – reserve in respect of claims notified to the insurer but not yet fully settled
 - Incurred but not enough reported – reserve for claims which have been reported but not all relevant details have been submitted to the insurer
 - Catastrophe reserve – amount of reserves held for abnormal events
 - Mismatch reserve – reserve for asset liability mismatch
 - Option and Guarantee reserve – reserve kept to meet the guarantees offered by insurer

(5)

[10 Marks]

Solution 6:

Some of the possible reasons for decline in solvency position:

New Business

- Type of Business - Company writes more capital intensive business (products where reserve requirement is higher) like IP and LTCI plans. Therefore strain from writing NB is higher for this company as opposed to the one with more PMI and CI business.
- Writing more guaranteed business, e.g. high sum assured plans, guaranteed premium plans, high replacement ratios in case of IP plans.
- Too passive Business strategy – writing little NB, therefore unable to recoup fixed expenses and overheads
- Too aggressive Business strategy - writing large NB volumes resulting in high strain
- Poor business mix sold - profile which has very low margin and high strain, e.g. policies with very high replacement ratio under IP plans

Claims

High claims outgo. Possible reasons:

- Poor claims management
- Poor underwriting standards. Underwriting not in line with what was assumed in pricing
- Increase in the incidence rates, resulting in more claims.
- Business mix sold significantly different from what was assumed in pricing
- Premium rate margin not sufficient
- More awareness/understanding of insurance products amongst policyholders, resulting in more claims
- One off incidence like pandemic
- Early diagnose of certain diseases resulting in higher outgoes
- Insufficient Reinsurance arrangement

Lapses

- Early lapses,
 - Asset share less than surrender value, resulting in financial loss for the company.
 - Premium received not sufficient to recover initial strain
- Selective lapses, worsening the future claim experience
- Too many lapses resulting in loss of future profits

Expenses

- Expenses higher than assumed in pricing
- Poor expense management, operational inefficiencies resulting in high fixed cost and overheads
- One off expenses like implementation of some project e.g. migration of data systems etc.
- Expense inflation higher than assumed
- Poor management decisions, e.g. launching too many products in market without doing the cost benefit analysis of the decision.

Market /Investment

- Credit quality of assets where investment made is poor. Investment in risky assets, thereby increasing the credit risk and solvency capital requirement of the company
- Too much concentration in a particular category of assets, resulting in increase in concentration risk
- Stock market crash
- Asset liability mismatch, therefore assets insufficient to meet future liabilities
- Change in the treatment of assets for solvency purpose by Regulator, for e.g. making bonds with credit rating lower than AA inadmissible for solvency purpose

Others

- Change in Regulatory regime, for e.g. from factor based approach to risk based approach increasing the solvency capital requirement
- Change in reserving methodology resulting in increase in reserve, reducing the free surplus available to support solvency
- Fine by Regulator on non-compliance, e.g. unethical selling practices
- Third party default, for e.g. default by Reinsurer
- Anti-selection/ fraud by the customers due to poor underwriting standard
- Loss of main distribution partner resulting in heavy loss of NB

- Guaranteed premium rates instead of reviewable premiums. Inability to reflect the increased risk in premiums

[12 Marks]

Solution 7:

i)

IP Plans – Guaranteed series of payments where payment amount can be:

1. Fixed
2. Escalated in line with a fixed rate or linked to an index (usually wage inflation)

CI Plans – Guaranteed in monetary terms. Fixed lump sum paid on legitimate claims irrespective of actual cost of treatment. Sometimes the lump-sum amount linked with index, e.g. medical inflation

PMI Plans – Usually indemnity based, amount of benefit paid to customer depends on cost of treatment – falls under the category of Indemnity. Depends on medical inflation

(3)

ii)

Purpose of Increase in exposure, whether it is only to maximize the return or

- to diversify as too much concentration in one type of asset increases concentration and default risk
- Replicating a benchmark portfolio etc.
- Regulatory constraints:
 - Whether the type of bonds under consideration allowed by Regulator
 - Whether increased exposure within Regulatory limits
- Impact on KPI's of the Company:
 - Company solvency position
 - Statutory Reserves – whether the amount invested is material enough to change long term interest rate assumption of the company, resulting in an impact on reserves.
 - Embedded value
- Cost versus Benefit analysis of the exercise,
 - Whether amount of asset considered for investment significant or minimal, is it worth doing the whole exercise
 - Return in relation to the risk involved, level of free assets available with the company in order to take that risk
 - Opportunity cost, if other better investments available
- Impact of this decision on the asset liability matching position of the Company. Whether the decision is in line with nature and term of products which Company writes
- If suitable corporate bonds available in the market,
 - E.g. bonds of longer duration to match the liabilities of long term plans like IP and CI.
 - Bonds with desired credit quality, for example AA rated and above

(5)

[8 Marks]

Solution 8:

i)

Difference in premium rates between Existing and Proposed plan due to:

Morbidity Experience

In Existing Products, both male and female lives are covered and unisex premium rates are offered to the customers. Unisex rates may have been arrived at by assuming a mix of male and female lives.

Premium rates vary between male and female lives due to difference in morbidity experience of the two lives – some of the reasons for variation listed below:

- Difference in occupation pattern of male and female. Risky jobs, e.g. mining usually undertaken by male lives
- Difference in smoking pattern, e.g. in certain societies, smoking is less common amongst female lives
- Difference in alcohol intake e.g. alcohol intake may be more amongst male life.
- Difference in dangerous hobbies, like participating in adventure games
- Difference in the diet or attitude towards exercise
- Biological differences in male and female bodies also contribute significantly towards different morbidity experience. Certain diseases are more prevalent in one gender e.g. certain forms of cancers like cervical cancer common in females

Proposed product covers only female lives and therefore, unisex premium rates may not be directly comparable with the premium rates offered under the proposed plan.

Claim conditions covered

The proposed plan only covers cancer whereas existing products cover wider range of CI diseases. The incidence rates under two product types may differ significantly, resulting in variation in premium rates.

E.g. - Existing products may also offer cover for Heart and Kidney diseases, which are completely excluded from proposed product. On the other hand, new product may be covering some forms of cancer, which are otherwise excluded from regular CI plans, e.g. Skin cancer.

Higher Margin for Adverse deviation

Proposed product would be the first product in market offering cover for diseases on standalone basis. Company may have built higher margin for adverse deviation in premium rates compared to existing products due to lack of experience.

Reinsurance Arrangement

Since there is no past experience for the proposed plan in market, Company would need significant help from the Reinsurer to price this product. Company may have opted for a lower retention limit under this plan compared to other products, resulting in higher premium pay-outs to the reinsurer. This cost would be baked in premium rates and would result in variation from existing rates.

Expenses

The proposed plan is first of its kind in the market and hence, greater marketing costs are associated in promoting this product. Also, the product would incur additional set-up related costs. A higher loading for expenses may be required in premiums under this product compared to existing plans.

(5)**ii)**

Possible rating factors:

- Age
- Occupation
- Location
- Benefit amount (sum assured)
- Smoker Status
- Medical history
- Alcohol habit
- Family's medical history
- Cooling off period/exclusions
- Types of cancer covered
- Policy Term
- Options provided, like tiered benefit, waiver of premium, indexed sum assured etc.
- BMI

(3)**iii)**

a. Some of the pricing challenges are:

Product design

- Difficult to ascertain the exact needs of customers and to create a product catering to those needs, for e.g. how much payment to be made at each stage of cancer. No similar product exists in the market, therefore, difficult to access the exact needs and create features addressing them.
- Deciding the types of options to be provided and doing a cost benefit analysis of those options. (If they add value to the company)
- Defining the policy wordings, terms and conditions, product exclusions etc.
- Making the product features consistent with the competitors. Too much deviation may make product difficult to sell.

Availability of data:

- Data available may not be sufficient enough to be divided into credible homogeneous groups like sum assured, occupation, location etc.
- Plan is offered only to female lives, which further reduces the level of data available.
- Data may not be available by various risk factors like types of cancer, progression of disease etc.
- Data available may not complete. There may be missing entries in the data
- Data may have errors
- Data available may be outdated for drawing any conclusive evidences. For e.g. breast cancer once life threatening, if detected in early stage is curable now.

Pricing Model and Assumptions

- Difficult to model complicated benefits
- Difficult to allow for flexible benefits like tiered benefits, waiver of premium etc. in pricing model
- Difficult to allow for guarantees and options in pricing model
- Requires multi state modeling in order to capture transition between various stages, for e.g. healthy state, different stages of cancer like CIS, early stage, critical stage etc.
- Medical inflation is challenging to assess
- Difficult to ascertain claim incidence rates with limited level of data

Underwriting

Significant morbidity risk associated with such plans and therefore, a strong underwriting policies is required

- Challenges in defining the approach to be taken for standard and substandard lives, e.g. whether to offer cover to a person who has family history of cancer, finalizing the medical grids basis which extra premium will be charged to sub- standard lives.
- Difficult to assess the level of underwriting to be done. Too strict and too relaxed underwriting process compared to the competitor poses a risk
- Difficult to do cost benefit analysis on the level of underwriting
- Challenges with the credibility of medical reports. Possibility of fraud from medical centers cannot be ruled out

Claim incidence and amount

- Assessing the rate of medical inflation a challenge
- With the new technological advances and diagnosis available,
 - Cancer is now usually identified at an early stage, thereby increasing the possibility of claim
 - Better medical facilities, increasing the life expectancy resulting in claim for longer duration
- Fraud, anti-selection by customers leading to higher claims payouts
- Difficult to access onset of cancer

Other challenges

Since no other product similar to this product exists in the market, there may be additional challenges associated with pricing, like:

- Regulatory constraints on product design, e.g. substandard rating factors, underwriting practices, exclusions etc
- No prior experience, therefore additional expenses associated in product set-up
 - May need to hire staff with prior experience in such products
 - Reinsurer assistance, which comes with a charge
 - May need separate claims and data management systems to handle such products
- There may be challenges associated with getting a comprehensive reinsurance arrangement for such plans
- May need to explore different distribution channels to reduce morbidity risk under such products e.g. online channel may have better morbidity experience than brokers.

(10)

iv)

Reserving challenges

- Data management is difficult. Policy systems will require major level of modifications in order to capture all the relevant fields required for reserving
- Models used for reserve estimation will require significant changes to model features like stage wise payouts, waiver of premiums etc.
- Monitoring of experience is difficult, e.g. analyzing the claims experience at different stages of cancer is a cumbersome exercise

As company has no prior experience in the product, it may need more frequent monitoring of various risk parameters. An extra level of prudence may be required in order to increase the level of confidence in reserve numbers.

(2)

[20 Marks]
